

Cross-Market Spillover Analysis Between Equity and Commodity Markets: A Comparative Study of USA and Brazil

Abstract

The increasing integration of global financial markets has intensified the transmission of shocks across asset classes, making the study of spillover effects crucial for portfolio management and risk assessment. This study investigates the return and volatility spillovers between equity markets and major commodity markets in two economically distinct countries: the United States and Brazil. While the United States represents a mature and highly developed financial system, Brazil serves as an emerging market with substantial dependence on commodity exports, particularly agricultural products.

Daily return data for equity indices, crude oil, gold, silver, and agricultural commodities are analyzed using a combination of econometric techniques, including the Augmented Dickey-Fuller (ADF) test, Vector Autoregression (VAR), Granger causality analysis, and GARCH-family volatility models. The findings reveal notable differences in the transmission mechanisms across the two economies. In the United States, equity market movements exhibit significant predictive power over precious metal returns, particularly gold and silver, indicating strong financial-market-driven spillovers. In contrast, Brazil demonstrates stronger linkages between equity markets and agricultural commodities, reflecting the structural importance of commodity production and exports within the Brazilian economy.

Volatility modelling further indicates persistent volatility clustering across all asset classes, with Brazilian equity markets exhibiting higher persistence than their US counterparts. These results suggest that the nature and intensity of spillovers are strongly influenced by a country's economic structure, with important implications for diversification strategies, hedging effectiveness, and financial stability.

Introduction

The growing interconnectedness of global financial markets has significantly increased the transmission of information, risk, and volatility across asset classes. Equity and commodity markets, which were traditionally viewed as relatively independent investment avenues, have become increasingly linked through globalization, cross-border capital flows, and changes in investor behaviour. As a result, shocks originating in one market may propagate to others, affecting asset prices, portfolio performance, and overall market stability.

Understanding these spillover mechanisms is particularly important for investors, policymakers, and risk managers seeking to design effective diversification and hedging strategies. Previous research has documented the existence of return and volatility spillovers between financial and

commodity markets; however, the magnitude and direction of these relationships often vary across countries depending on their economic structures and levels of market development.

This study compares the United States and Brazil, two economies with fundamentally different characteristics. The United States represents one of the world's largest and most sophisticated financial markets, whereas Brazil is a major commodity-exporting emerging economy whose economic performance is closely tied to agricultural and resource-based sectors. By comparing these two countries, the study seeks to determine whether market structure influences the transmission of shocks between equity and commodity markets.

The primary objectives of the research are:

- To investigate the existence of return spillovers between equity and commodity markets.
- To identify the direction and strength of these spillovers.
- To examine the persistence of volatility shocks across different asset classes.
- To compare spillover dynamics between a developed economy and a commodity-dependent emerging market.

Data and Variables

The empirical analysis utilizes daily price data covering five major asset categories:

1. Equity Market Index
2. Crude Oil
3. Gold
4. Silver
5. Agricultural Commodities

To eliminate non-stationarity commonly observed in financial price series, logarithmic returns are computed as:

$$R(t) = \ln(P(t)/P(t-1))$$

where $P(t)$ represents the current price and $P(t-1)$ represents the previous period price.

The use of return series rather than price levels is standard practice in financial econometrics because returns generally exhibit more stable statistical properties and satisfy the stationarity requirements of time-series modelling techniques.

Methodology

A multi-stage econometric framework is employed to capture both return transmission and volatility dynamics across markets.

Stationarity Analysis

The Augmented Dickey-Fuller (ADF) test is first applied to determine whether the return series are stationary. Establishing stationarity is a prerequisite for reliable estimation of VAR models and causality tests.

Equity ADF statistic = -18.0557

Oil ADF statistic = -11.9694

Gold ADF statistic = -30.5897

Silver ADF statistic = -28.8956

Agro ADF statistic = -15.9339

All variables had p-values equal to 0.000.

Therefore, all series were stationary.

For Brazil:

Equity ADF statistic = -15.9181

Oil ADF statistic = -12.2295

Gold ADF statistic = -30.4956

Silver ADF statistic = -28.4210

Agro ADF statistic = -28.0420

Again, all variables were stationary with p-values equal to 0.000.

Spillover Analysis Using VAR

A Vector Autoregression (VAR) framework is employed to examine dynamic interactions among asset returns. Unlike single-equation models, VAR treats all variables as endogenous, allowing each market to be influenced by both its own past values and the lagged values of other markets. VAR(1) was estimated.

1.1 USA Results

The USA VAR results indicate that equity markets have important effects on precious metals.

In the Gold equation:

Previous equity returns had a coefficient of 0.0781 with a p-value of 0.008. This indicates a statistically significant relationship between equity movements and gold returns.

In the Silver equation:

Previous equity returns had a coefficient of 0.1153 with a p-value of 0.041. This also indicates significant equity-to-silver transmission.

The results suggest that precious metals in the US market are not completely independent from equity market movements.

Granger Causality Testing

To identify the direction of information transmission, Granger causality tests are conducted. These tests determine whether past movements in one asset contain statistically significant information for predicting future movements in another asset.

Significant relationships were:

Equity $\rightarrow$ Gold

p-value = 0.0008

This indicates that previous equity movements help predict gold returns.

Equity $\rightarrow$ Silver

p-value = 0.0298

This indicates equity market information affects silver.

Silver $\rightarrow$ Gold

p-value = 0.0442

This indicates silver movements influence gold.

Agro $\rightarrow$ Silver

p-value = 0.0474

This indicates agricultural commodity movements affect silver.

Overall, the USA shows stronger financial market driven spillovers.

1.2 USA Volatility Results

GARCH(1,1) models volatility persistence.

The model is:

Variance = constant + ARCH effect + GARCH effect

The ARCH coefficient measures immediate reaction to shocks.

The GARCH coefficient measures how long volatility persists.

USA Equity:

Alpha = 0.0983

Beta = 0.8853

The high beta value shows that volatility shocks continue to affect future volatility.

1.3 USA Gold:

Beta = 0.9174

This indicates strong volatility persistence.

EGARCH results show asymmetric volatility effects.

For equity:

Alpha = 0.2083

This suggests negative shocks have a stronger impact on volatility.

2.1 Brazil Results:

Brazil shows a different spillover pattern compared to the USA.

The strongest relationship was between equity markets and agricultural commodities.

Equity → Agro:

Coefficient = 0.0699

p-value = 0.028

This shows significant transmission from equity markets into agricultural commodities.

Gold negatively influenced agricultural commodities.

Coefficient = -0.1078

p-value = 0.031

Silver positively influenced agricultural commodities.

Coefficient = 0.0573

p-value = 0.033

This reflects Brazil's commodity-dependent economic structure.

2.2 Brazil Granger Causality Results

Significant spillovers:

Equity → Agro

p-value = 0.0167

This shows that equity market movements help predict agricultural commodity movements.

Silver → Gold

p-value = 0.0139

This indicates transmission between precious metals.

Compared to the USA, Brazil shows weaker equity-to-precious metal spillovers and stronger commodity-related relationships.

2.3 Brazil Volatility Results

Brazil equity volatility:

Alpha = 0.0253

Beta = 0.9710

The beta value is higher than the USA.

USA equity beta:

0.8853

Brazil equity beta:

0.9710

This means volatility shocks persist longer in Brazil.

Gold volatility:

Beta = 0.9183

Agricultural commodities:

Beta = 0.8702

All assets show significant volatility persistence.

Comparative Discussion

The empirical findings reveal distinct spillover structures across the two economies.

In the United States, significant transmission effects are primarily observed between equity markets and precious metals. The results indicate that movements in the stock market contain predictive information for both gold and silver returns. This suggests that precious metals may not function as fully independent diversification instruments during periods of market stress, as they remain influenced by developments in broader financial markets.

In contrast, Brazil exhibits stronger interactions between equity markets and agricultural commodities. Given the substantial contribution of agricultural exports to Brazil's economy, changes in domestic financial conditions appear to be more closely linked to agricultural market performance. This highlights the importance of economic structure in shaping cross-market dependencies.

Furthermore, volatility persistence is considerably stronger in Brazil than in the United States. The higher GARCH persistence parameter observed for Brazilian equities indicates that volatility shocks dissipate more slowly, potentially increasing the duration of market uncertainty following major economic events.

Economic Implications

The findings carry important implications for both investors and policymakers.

From an investment perspective, the results suggest that diversification strategies should account for country-specific spillover dynamics rather than assuming uniform relationships across markets. Assets traditionally considered safe havens, such as gold and silver, may still be affected by equity market developments, particularly in developed economies.

For policymakers, understanding spillover channels can improve financial stability monitoring by identifying markets that may serve as transmission mechanisms during periods of economic stress. In commodity-dependent economies such as Brazil, disruptions in commodity markets may have broader consequences for financial market performance and economic growth.

Limitations

The study has limitations:

1. VAR captures linear relationships only.
2. The sample period may not capture long-term cycles.
3. GARCH models capture volatility persistence but not all nonlinear dependencies.
4. Extreme events may require additional frameworks.

Conclusion

This study analysed return and volatility spillovers between equity and commodity markets in the USA and Brazil. The results show that spillovers exist but differ across economies. The USA exhibits stronger equity-driven spillovers towards precious metals, while Brazil shows stronger equity-agriculture connections. Volatility models confirm persistent shocks in both markets, with Brazil showing greater persistence in equity volatility. The findings highlight the importance of considering economic structure when analysing diversification and risk management strategies.

Appendix

```
import yfinance as yf
import pandas as pd
import numpy as np
```

```

from statsmodels.tsa.stattools import adfuller, grangercausalitytests
from statsmodels.tsa.api import VAR
from arch import arch_model
COUNTRY = "USA"
if COUNTRY == "BRAZIL":
    tickers = {
        "EQUITY": "^BVSP",
        "OIL": "CL=F",
        "GOLD": "GC=F",
        "SILVER": "SI=F",
        "AGRO": "DBA"
    }
else:
    tickers = {
        "EQUITY": "^GSPC",
        "OIL": "CL=F",
        "GOLD": "GC=F",
        "SILVER": "SI=F",
        "AGRO": "DBA"
    }
start = "2022-01-01"
end = "2025-06-01"
print("Downloading data...")
prices = pd.DataFrame()
for name,ticker in tickers.items():
    data = yf.download(
        ticker,
        start=start,
        end=end,
        auto_adjust=True,
        progress=False
    )
    prices[name] = data["Close"]
prices = prices.dropna()
returns = np.log(
    prices / prices.shift(1)
)
returns = returns.dropna()
returns.to_csv(
    COUNTRY+"_returns.csv"
)
print("\nReturns created")
print(returns.head())
print("ADF TEST")

```

```

for col in returns.columns:
    result = adfuller(
        returns[col]
    )
    print(
        col,
        "ADF:",
        round(result[0],4),
        "p-value:",
        round(result[1],4)
    )
print("VAR MODEL")
model = VAR(returns)
lag = model.select_order(10)
print(lag.summary())
var = model.fit(1)
print(var.summary())
print("GRANGER CAUSALITY")
cols = returns.columns
for cause in cols:
    for effect in cols:
        if cause != effect:
            print(
                "\n",
                cause,
                "->",
                effect
            )
            test = grangercausalitytests(
                returns[[effect,cause]],
                maxlag=2,
                verbose=False
            )
            pvalue = test[2][0]['ssr_ftest'][1]
            print(
                "p-value:",
                round(pvalue,4)
            )
            if pvalue < 0.05:
                print("SIGNIFICANT SPILLOVER")
            else:
                print("No significant spillover")
print("GARCH / EGARCH")
for col in returns.columns:

```

```
print("\nAsset:",col)
data = returns[col]*100
# GARCH(1,1)
garch = arch_model(
    data,
    vol="GARCH",
    p=1,
    q=1
)
garch_result = garch.fit(
    disp="off"
)
print("\nGARCH(1,1)")
print(garch_result.summary())
# EGARCH
egarch = arch_model(
    data,
    vol="EGARCH",
    p=1,
    q=1
)
egarch_result = egarch.fit(
    disp="off"
)
print("\nEGARCH(1,1)")
print(egarch_result.summary())
print("DONE")
```